Unit 2 FRQ Set 1: Coaching Guide

Use this in two passes. **Pass 1:** for any part you couldn't finish, read only the 'Coach' questions and try again (a different colour pen). **Pass 2:** grade both attempts against the rubric.

Question	Max	Attempt 1 (timed)	After coaching
Q1 MR	10		
Q2 TBR	12		
Q3 LAB	10		
Q4 QQT	8		
Total	**40**		

Rubrics follow the College Board's style ('1 point for…'). They're written for practice. They aren't official scoring guidelines, but they test the same skills.

Question 1: Mathematical Routines

10 points

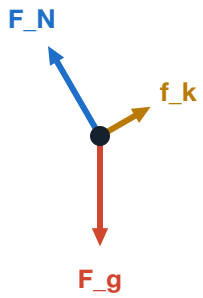
A block of mass m is released from rest at the top of a ramp of length D that makes an angle θ with the horizontal. The coefficient of kinetic friction between the block and the ramp is μ_k.

Part A (2 pts)

COACH: ASK YOURSELF

- Which objects touch the block? Which object pulls on it without touching?
- Which way is the block sliding? Friction opposes that.

A FULL-CREDIT ANSWER



Three forces: F_g straight down, F_N perpendicular to the ramp surface, and kinetic friction up the slope, opposite the sliding.

Correct forces only: gravity down, normal perpendicular to the ramp, friction up the ramp	1 pt
Labels clear, arrows start on the dot, no extra or 'component' arrows	1 pt

WHERE POINTS LEAK Drawing F_N straight up (it must be perpendicular to the surface), adding a 'down-the-ramp force' (that's just part of gravity), or drawing the components as separate forces.

Part B (3 pts)

COACH: ASK YOURSELF

- Tilted axes: along the ramp, which forces (or parts of forces) act?
- What is F_N here? Is it mg? (It isn't. Get it from the perpendicular direction.)
- The question doesn't list m. What does that tell you is going to happen?

A FULL-CREDIT ANSWER

`ΣF = ma` along the ramp, with + pointing down the slope.

Perpendicular: $F_N - mg\cos\theta = 0 \rightarrow F_N = mg\cos\theta$.

Along: $mg\sin\theta - \mu_k mg\cos\theta = ma$.

$a = g(\sin\theta - \mu_k \cos\theta)$

Starts from Newton's 2nd law along the incline	1 pt
Correct normal force mg cos θ, used in the friction $\mu_k F_N$	1 pt
Correct final answer with m eliminated, in the requested variables	1 pt

WHERE POINTS LEAK Writing $\mu_k mg$ (forgetting cos θ), leaving m in the answer when the question didn't list it, or jumping straight to the answer without the starting equation.

Part C (2 pts)

COACH: ASK YOURSELF

- Plug into part B. Do you need m? (No.)
- Released from rest, with a known distance and constant a. Which kinematics equation has no final velocity in it?

A FULL-CREDIT ANSWER

(i) $a = 9.8(\sin 30° - 0.20\cos 30°) = 9.8(0.500 - 0.173) =$ **3.2 m/s²**.

(ii) $D = \frac{1}{2}at^2 \rightarrow t = \sqrt{(2D/a)} = \sqrt{(2\cdot 2.0/3.20)} =$ **1.1 s**.

Correct acceleration with units (consistent with part B)	1 pt
Correct time using a kinematics equation	1 pt

WHERE POINTS LEAK Calculator in radian mode (sin 30 = –0.99 in radians!). Check that sin 30° = 0.5 before the test.

Part D (3 pts)

COACH: ASK YOURSELF

- The student is partly right. Which force really IS bigger for the heavier block?
- What else gets bigger too? Look at your expression from part B.

A FULL-CREDIT ANSWER

Disagree. The heavier block does have a larger gravitational component down the ramp, but its friction force ($\mu_k mg \cos\theta$) is also twice as big, and it has twice the inertia. From part B, $a = g(\sin\theta - \mu_k \cos\theta)$ does not depend on m. The same acceleration over the same distance from rest gives the same time.

Disagree (or 'same time'), with an attempted justification	1 pt
Refers to the derived expression (or ΣF and m both scaling) to show a is independent of mass	1 pt
Connects equal acceleration and the same D to an equal time	1 pt

WHERE POINTS LEAK Just saying 'mass doesn't matter' with no reason. Graders want the specific link: the force doubles AND the mass doubles, so a = F/m is unchanged.

THINK FURTHER What angle makes the block slide at constant speed? ($\tan\theta = \mu_k$, so $\theta \approx 11°$.) If θ were less than that, what would the block do after a small push? (It slows down and stops.)

Question 2: Translation Between Representations

12 points

Block 1 (mass M) and Block 2 (mass 3M) hang from opposite ends of a light string that passes over an ideal (massless, frictionless) pulley. The blocks are released from rest.

Part A (3 pts)

COACH: ASK YOURSELF

- Each block has only two forces. But which arrow is longer on each block? Which way does each block accelerate?
- Is the string tension the same on both blocks? (Light string, ideal pulley.)

A FULL-CREDIT ANSWER



Block 1: T longer than Mg (it accelerates up). Block 2: 3Mg longer than T (it accelerates down). Both T arrows the same length. 3Mg is three times Mg.

Each diagram has exactly tension up and gravity down	1 pt
Block 1: T > Mg; Block 2: 3Mg > T	1 pt
Same T on both diagrams; 3Mg drawn about 3 times Mg	1 pt

WHERE POINTS LEAK Equal arrows on a block (that would mean no acceleration), or different tensions on the two blocks.

Part B (4 pts)

COACH: ASK YOURSELF

- Fastest route to a: treat both blocks as one system along the string. What's the net driving force? What's the total mass?
- Then isolate ONE block (either one) to find T.

A FULL-CREDIT ANSWER

System along the string: $\Sigma F = ma$ → $3Mg - Mg = (M + 3M)a$ → **$a = g/2$**.

Block 1 (up +): $T - Mg = Ma$ → $T = M(g + g/2) =$ **$(3/2)Mg$**.

Check with Block 2 (down +): $3Mg - T = 3M(g/2)$ → $T = 3Mg - (3/2)Mg = (3/2)Mg$ ✓

Applies Newton's 2nd law to a system or to individual blocks with consistent signs	1 pt
Correct a = g/2	1 pt

Applies Newton's 2nd law to one block to find T	1 pt
Correct T = 3Mg/2	1 pt

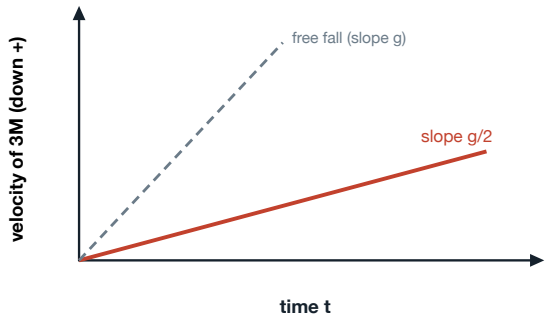
WHERE POINTS LEAK Sign errors from using 'up = +' for BOTH blocks. Pick + as the direction each block moves, and the signs take care of themselves.

Part C (3 pts)

COACH: ASK YOURSELF

- Is the acceleration constant? Then what shape is v vs. t?
- Slope = acceleration. Compare g/2 with g.

A FULL-CREDIT ANSWER



Block 2: straight line starting at the origin	1 pt
Positive slope (downward positive)	1 pt
Free-fall line steeper, about twice the slope, also from the origin	1 pt

WHERE POINTS LEAK Drawing a curve, starting above zero (it starts at rest), or drawing the two lines with the same slope.

Part D (2 pts)

COACH: ASK YOURSELF

- In part A you drew T bigger than one weight and smaller than the other. Does (3/2)Mg fit between Mg and 3Mg?

A FULL-CREDIT ANSWER

Yes. In part A, T had to be greater than Mg (Block 1 accelerates up) and less than 3Mg (Block 2 accelerates down). T = 1.5Mg lies between Mg and 3Mg, so it's consistent with both diagrams.

Says the diagrams require Mg < T < 3Mg, using the direction of acceleration of each block	1 pt
Shows 1.5Mg satisfies both inequalities	1 pt

WHERE POINTS LEAK Just writing 'yes, it matches'. You need to name WHAT matches: the inequality, and the reason for it.

THINK FURTHER Replace Block 2 with a mass n·M. Show that $a = g(n - 1)/(n + 1)$. What does a approach as $n \to \infty$? (g. Block 2 is basically in free fall.)

Question 3: Experimental Design and Analysis

10 points

Students want to determine the coefficient of kinetic friction μ_k between a wooden block and a lab table. They have the block, a set of masses that can be stacked on top of it, a spring scale, a balance, a meterstick and a stopwatch.

Part A (2 pts)

COACH: ASK YOURSELF

- Friction equals the spring scale reading only under one condition. Which condition? (Think $\Sigma F = 0$.)
- What will you vary, so that you get a graph and not just one number?

A FULL-CREDIT ANSWER

1. Measure the mass of the block with the balance, then add masses on top to change the total mass (at least 5 different values).
2. For each mass, pull the block horizontally with the spring scale so it slides at a **constant speed**, and read the scale. At constant velocity, $\Sigma F = 0$, so the scale reading equals the friction force.
3. Repeat each trial 3 times and average the readings. Keep the scale horizontal.

Varies the normal force (added mass) and measures the friction force with the spring scale	1 pt
Pulls at constant velocity (so the reading = friction), plus repeated trials or a horizontal pull	1 pt

WHERE POINTS LEAK Not saying 'constant velocity'. Without it, the scale reading isn't the friction force, and the method fails.

Part B (2 pts)

COACH: ASK YOURSELF

- On a level table with a horizontal pull, F_N = ?
- $f = \mu_k F_N$ looks like $y = mx$. Which variable is y, and which is x?

A FULL-CREDIT ANSWER

$F_N = mg$: 4.9, 9.8, 14.7, 19.6, 24.5 N.

Graph **friction force (vertical)** vs. **normal force (horizontal)**. Since $f = \mu_k F_N$, the slope is μ_k.

Correct normal forces (mg)	1 pt
Graphs f vs. F_N (or an equivalent linearization) and says slope = μ_k	1 pt

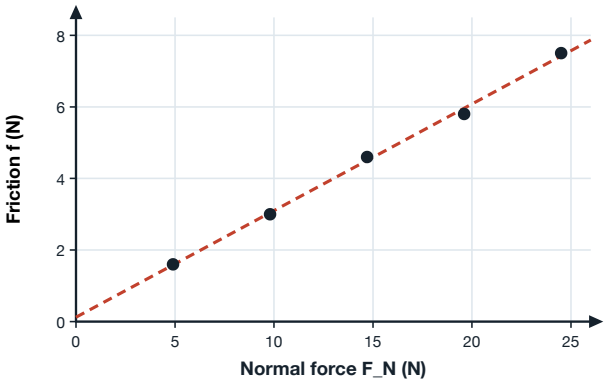
WHERE POINTS LEAK Graphing f vs. mass. That works, but then the slope is $\mu_k g$, and you have to divide by g in part D. Say so if you do it.

Part C (4 pts)

COACH: ASK YOURSELF

- Pick a scale that uses more than half the grid in each direction.
- A best-fit line is NOT connect-the-dots, and it doesn't have to pass through every point.

A FULL-CREDIT ANSWER



Axes labelled with quantities and units	1 pt
Linear scale that uses most of the grid	1 pt
Points plotted correctly	1 pt
Straight best-fit line with points scattered on both sides	1 pt

WHERE POINTS LEAK A scale crammed into one corner, connecting the dots, or forcing the line through the first and last points.

Part D (2 pts)

COACH: ASK YOURSELF

- Pick two points ON YOUR LINE (not data points), far apart.
- Does μ have units? (No. N/N cancels.)

A FULL-CREDIT ANSWER

Slope = $\Delta f / \Delta F_N$, using two points on the line, for example (0, 0.12) and (25, 7.57): slope ≈ **0.30**, so $\mu_k \approx$ **0.30** (no units). Anything from about 0.28 to 0.32 is consistent with the data.

Uses two points from the best-fit line (not data points) to get the slope	1 pt
$\mu_k \approx 0.30$ with no units, consistent with the student's line	1 pt

WHERE POINTS LEAK Picking two data points instead of points on the line. The rubric specifically checks for this.

THINK FURTHER Why do the students repeat each trial? Name one source of error that repeating DOESN'T fix. (Systematic error: for example, a spring scale that reads 0.2 N too high. That shifts the intercept, but notice it doesn't change the slope!)

Question 4: Qualitative/Quantitative Translation

8 points

Two identical cars drive around the same flat, unbanked curve of radius R. Car 2 travels at twice the speed of Car 1. Both cars make the turn without skidding.

Part A (3 pts)

COACH: ASK YOURSELF

- What force turns the car? What does 'not skidding' tell you about which kind of friction it is?
- A faster car on the same curve changes the direction of its velocity faster AND has more speed to redirect. Does it need more or less sideways force?

A FULL-CREDIT ANSWER

Static friction from the road is the only horizontal force on each car, so it is what pulls the car toward the center of the curve. Car 2 moves faster, so its velocity has to change direction more quickly, and each change is bigger. It needs a **larger acceleration toward the center**, and so a larger friction force. The cars are identical, so their normal forces are equal, and Car 2 therefore needs a **greater coefficient of static friction** than Car 1.

Identifies static friction as the force toward the center of the curve	1 pt
Says the faster car needs a larger centripetal acceleration/force	1 pt
Same normal force → Car 2 needs a larger μ_s	1 pt

WHERE POINTS LEAK Writing an equation in a 'without equations' part loses the reasoning points. Use words that describe the physics.

Part B (3 pts)

COACH: ASK YOURSELF

- Which direction is 'toward the center'? Write $\Sigma F_c = mv^2/R$.
- 'Minimum μ' means friction is at its maximum: $f = \mu F_N$.

A FULL-CREDIT ANSWER

Toward the center: $\Sigma F = ma_c$ → $f_s = mv^2/R$.

Vertical: $F_N = mg$. At the minimum μ, $f_s = \mu_{min} F_N = \mu_{min} mg$.

$\mu_{min} mg = mv^2/R$ → $\mathbf{\mu_{min} = v^2/(gR)}$

Newton's 2nd law with centripetal acceleration v^2/R	1 pt
Static friction = μF_N with $F_N = mg$	1 pt
Correct expression	1 pt

WHERE POINTS LEAK Adding a separate 'centripetal force' to the forces. Friction IS the centripetal force here.

Part C (2 pts)

COACH: ASK YOURSELF

- μ_{min} depends on v how? Linearly or squared?

A FULL-CREDIT ANSWER

$\mu_{min} = v^2/(gR)$ increases with v, which matches part A: the faster car needs more friction. Because $\mu_{min} \propto v^2$, doubling the speed makes Car 2 need **4 times** the coefficient of Car 1.

Says the expression increases with v, consistent with part A	1 pt
Factor of 4 because of v^2	1 pt

WHERE POINTS LEAK Saying 'twice', which forgets the square. This is exactly the factor-of-change skill.

THINK FURTHER Why are highway curves banked? Draw the FBD of a car on a banked curve with no friction. Which force now has a component toward the center? (The normal force.)

Error log: FRQ Set 1

For each miss or guess: what went wrong, using one of these types: **C** concept (didn't know the physics) · **R** representation (FBD/graph wrong) · **M** math/algebra · **F** FRQ wording (right idea, missing the words) · **T** time.

Q	Type	What I did vs. what I should have done	Re-try date / ✓

Score: _____ / 40 Sure-and-right: _____ Time used: _____